

Lab Experiments 13-15

Data Handling & Visualization: Python Code & Output

Experiment 13

Word Frequency Distribution from a Text Document

PYTHON

Code

```
import string
from collections import Counter

# Read the text document
with open("sample_text.txt", "r") as file:
    text = file.read()

# Preprocess: lowercase and remove punctuation
text = text.lower()
text = text.translate(str.maketrans('', '', string.punctuation))

# Split into words and count frequency
words = text.split()
word_freq = Counter(words)

# Display the frequency distribution (most common first)
print("Word Frequency Distribution:")
for word, freq in word_freq.most_common():
    print(f"{word}: {freq}")
```

Output

```
Word Frequency Distribution:
data: 12
and: 8
science: 5
to: 4
is: 3
good: 3
uses: 2
insights: 2
from: 2
the: 2
of: 2
an: 1
interdisciplinary: 1
field: 1
that: 1
scientific: 1
methods: 1
processes: 1
algorithms: 1
systems: 1
extract: 1
knowledge: 1
structured: 1
unstructured: 1
```

```
related: 1
mining: 1
machine: 1
learning: 1
big: 1
techniques: 1
theories: 1
drawn: 1
many: 1
fields: 1
within: 1
context: 1
mathematics: 1
statistics: 1
computer: 1
information: 1
a: 1
scientist: 1
collects: 1
cleans: 1
analyzes: 1
builds: 1
models: 1
help: 1
organizations: 1
make: 1
better: 1
decisions: 1
foundation: 1
every: 1
decision: 1
leads: 1
```

Experiment 14

Frequency Distribution of Customer Ages

PYTHON

Code

```
import pandas as pd

# Sales data for the past month, stored in a Pandas DataFrame
data = {
    'CustomerID': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12],
    'Age': [25, 34, 25, 41, 34, 28, 25, 34, 41, 30, 28, 25],
    'Product': ['A', 'B', 'A', 'C', 'B', 'A', 'C', 'B', 'A', 'C', 'B', 'A'],
    'Amount': [500, 750, 500, 1200, 750, 450, 1200, 750, 500, 900, 450, 500]
}
df = pd.DataFrame(data)

# Frequency distribution of customer ages
age_freq = df['Age'].value_counts().sort_index()

print("Frequency Distribution of Customer Ages:")
print(age_freq)
```

Output

Frequency Distribution of Customer Ages:

```
Age
25    4
28    2
30    1
34    3
41    2
Name: count, dtype: int64
```

Experiment 15

Frequency Distribution of Likes Among Posts

PYTHON

Code

```
import pandas as pd

# User interaction data: number of likes received by each post
data = {
    'PostID': list(range(1, 13)),
    'Likes': [120, 85, 120, 200, 85, 45, 120, 200, 85, 60, 45, 120]
}
df = pd.DataFrame(data)

# Frequency distribution of likes among the posts
likes_freq = df['Likes'].value_counts().sort_index()

print("Frequency Distribution of Likes Among Posts:")
print(likes_freq)
```

Output

```
Frequency Distribution of Likes Among Posts:
Likes
45    2
60    1
85    3
120   4
200   2
Name: count, dtype: int64
```